

Recitation 4

ECON UN3412: Econometrics

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1 Joint Significant Testing

Consider a liner model with 3 explanatory variables:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i. \quad (1)$$

Suppose we want to test a null hypothesis involving multiple coefficients at once, for example

$$\begin{aligned} H_0 : \beta_1 = \beta_2 = 0 \\ H_1 : \beta_1 \neq 0 \text{ or } \beta_2 \neq 0 \end{aligned} \quad (2)$$

We will use an F -test for this purpose.

1.1 F-test with homoskedastic errors

Consider a regression where we impose the restriction that $\beta_1 = \beta_2 = 0$. The *restricted* regression is

$$Y_i = \beta_0 + \beta_3 X_{3i} + u_i \quad (3)$$

The *unrestricted* regression is given by Equation 1. The F statistic is given by

$$F = \frac{R_{unres}^2 - R_{res}^2}{q} / \frac{1 - R_{unres}^2}{n - k - 1} \quad (4)$$

where R_{unres}^2 and R_{res}^2 are the R-squared values from the unrestricted and restricted regressions, respectively, and $q = 2$ is the number of restrictions, n is the sample size, and $k = 3$ is the number of regressors in the unrestricted regression (*excluding* the intercept).

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When both multiple coefficients are jointly significant, it motivates keeping both variables in the regression, despite potential concerns of multicollinearity inflating standard errors.

2 Testing Single Restrictions on Multiple Parameters

Now consider from Equation 1 we want to test the following:

$$\begin{aligned} H_0 : \beta_1 &= \beta_2 \\ H_1 : \beta_1 &\neq \beta_2 \end{aligned} \tag{5}$$

To test this hypothesis, we can rearrange it as

$$\begin{aligned} H_0 : \beta_1 - \beta_2 &= 0 \\ H_1 : \beta_1 - \beta_2 &\neq 0 \end{aligned} \tag{6}$$

We can test this with a standard t-statistic:

$$t = \frac{\hat{\beta}_1 - \hat{\beta}_2}{se(\hat{\beta}_1 - \hat{\beta}_2)} \tag{7}$$

Where $SE(\hat{\beta}_1 - \hat{\beta}_2) = \sqrt{\text{Var}(\hat{\beta}_1) + \text{Var}(\hat{\beta}_2) - 2\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)}$. This requires the covariance term $\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)$, which is not typically reported in standard regression tables.

The Reparameterization Trick

Instead of calculating the covariance by hand, we can rewrite the regression so that software directly estimates $\gamma_1 = \beta_1 - \beta_2$ along with its standard error.

1. Define the parameter of interest:

$$\gamma_1 = \beta_1 - \beta_2 \implies \beta_1 = \gamma_1 + \beta_2$$

2. Substitute $\beta_1 = \gamma_1 + \beta_2$ into the original model (Equation 1):

$$\begin{aligned} Y_i &= \beta_0 + (\gamma_1 + \beta_2)X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i \\ &= \beta_0 + \gamma_1 X_{1i} + \beta_2 (X_{1i} + X_{2i}) + \beta_3 X_{3i} + u_i \end{aligned}$$

3. Define the new regressor $W_i = X_{1i} + X_{2i}$. Note that we replace X_{2i} with W_i

$$Y_i = \beta_0 + \gamma_1 X_{1i} + \beta_2 W_i + \beta_3 X_{3i} + u_i \tag{8}$$

Now:

- The OLS coefficient on X_{1i} is $\hat{\gamma}_1 = \hat{\beta}_1 - \hat{\beta}_2$.
- The standard error reported by Stata for X_{1i} is $\text{SE}(\hat{\gamma}_1) = \text{SE}(\hat{\beta}_1 - \hat{\beta}_2)$.
- Testing $H_0 : \beta_1 = \beta_2$ simplifies to testing $H_0 : \gamma_1 = 0$ using the routine t -statistic on X_{1i} .